BATCH8_FIBRE_RR: Residual fibre law vs 5n²/2 theta mass mod 8

2026-10-24 (America/Chicago)

RESULTS: BATCH8_FIBRE_RR (Map-FRR1)

PREREG.md sha256 b75fa7dadfea3c1ed223bf8c25b51d8b68524841887a3db42b50b7dc62947c2d. Hashed after exploration (data/explore.sha256) and before scripts/test.py ran (logs/test.log, data/test.json). Authorization: Arvin's "Build" (2026-10-24, America/Chicago), for this map only.

fibre	ρ_res (Residual, Cat. A)	ρ_res_mod (170n² mod 8)
0	1/32	1/2
1	1/4	0
2	3/32	1/2
3	0	0
4	7/32	0
5	1/4	0
6	5/32	0
7	0	0

- P1 (exact equality): **false**. TV = 7/8.
- Supports: Resonant {0,2} vs Residual {0,1,2,4,5,6}.
- Zeros on {3,7}: yes. But that is an **accident** of the square-class obstruction, not an independent reason; fibres 1,4,5,6 also vanish.
- Non-vacuity: the committed map is not uniform on the six open fibres. The test is non-vacuous.

Verdict: FAIL (Category A: exact rational arithmetic). Bridge: **NO.** No Residual ↔ Resonant claim. ## NOTES: BATCH8_FIBRE_RR ## Explore candidates (data/explore.json; modular sources only) | candidate | 8-tuple on Z/8 | |—|—| | 5n²/2 ×68 = 170n² (COMMITTED) | (1/2,0,1/2,0,0,0,0,0) | | R²/34 ×68 = 2R² | (1/2,0,1/2,0,0,0,0,0) | | 35m²/4 ×68 = 595m² | (1/4,0,0,1/2,1/4,0,0,0) | | pooled three (×68) | (5/12,0,1/3,1/6,1/12,0,0,0) | | 5n² (min. scaling) | (1/4,0,0,0,1/4,1/2,0,0) | | R² (min.) | (1/4,1/2,0,0,1/4,0,0,0) | | 35m² (min.) | (1/4,0,0,1/2,1/4,0,0,0) | | abs χ_140 on n mod 8 | (0,1/4,0,1/4,0,1/4,0,1/4) | | signed χ_140 on n mod 8 (period 280) | (0,…,0), identically

zero | Not computed: a mod-8 indexing of the 105 companions / T-phases k/9800 (I found no written mod-8 indexing), and the α1-shifted characteristic u ∈ α1+Z (α1 is not fixed numerically in the files read).

Lemma S (Category A, elementary): square-class obstruction

For any c, the values c·n² mod 8 lie in c·{0,1,4}, so there are at most 3 fibres. ρ_res has 6 non-zero fibres. Therefore **no** unary-theta exponent mass (any scaling, any of the three thetas, with integer n ∈ Z) can equal ρ_res. Mixtures of the three ×68 thetas cover only {0,2,3,4}, which misses 1, 5 and 6. abs χ_140 is supported on the odd classes, with 3 and 7 non-zero. So **every** explored written candidate is excluded. The FAIL is structural, not a near miss. ## Implication Any Resonant source for ρ_res would need a 2-adic object with mass on 6 classes and zeros at 3 and 7, i.e. a weight beyond the n² square classes (e.g. an α1-shifted characteristic or a written companion indexing). Building one now would be inventing objects, so it is not done here.